Total No. of Pages-3 *Roll No.* 2K21/B2/52

Vth Semester

B.Tech.

End Term Examination **Nov./Dec.- 2023**

HU307 Basic Econometrics

Time- 3.00 hrs **Marks- 50**

Note: 1. Answer any five questions.
2. All questions carry equal marks.
3. Assume missing value, if any

1 a What do you mean by Econometrics? Explain its methodology with a suitable example. [5] [CO#1]

b Decadal percentage growth of urban population in India is given below [2] [CO#1]

Years	1951	1961	1971	1981	1991	2001	2011
Decadal percent increase	8.25	19.08	32.09	41.49	25.85	37.91	46.02

Find out percentage growth rate of urban population during these seven decades.

c You are given the following incomplete frequency distribution. It is known that the total frequency is 1000 and the median is 413.11. Estimate by calculation the missing frequencies and find the value of mode. [3] [CO#1]

Value	300-325	325-350	350-375	375-400	400-425	425-450	450-475	475-500
Frequency	5	17	80	?	326	?	88	9

2 a In hypothesis testing, which one is your research question, Null Hypothesis or Alternative Hypothesis? Which hypothesis you reject or could not reject? Discuss with suitable examples. [5] [CO#2]

b For a distribution of 280 observations, mean and standard deviation were found to be 54 and 3 respectively. On checking it was found that two observations which should correctly read as 62 and 82 and has been wrongly recorded as 64 and 80 respectively. Calculate the correct value of mean and standard deviation. [2] [CO#1]

c For a group containing 50 male workers, the mean and standard deviation of their weekly wages are Rs. 63 and R. 9 respectively. For a group of 40 female workers, these are Rs. 54 and Rs. 6 respectively. Find the standard deviation for the combined group of 90 workers. [3] [CO#1]

3 a You have to find out variables which significantly effects students' preference to study engineering. How you will find them? Discuss stepwise . [5] [CO#2]

b Calculate Karl Pearson's coefficient of correlation between price and supply of a commodity from the following data: [5] [CO#2]

Price (in Rs)	17	18	19	20	21	22	23	24	25	26
Supply (in Kg)	38	37	38	33	32	33	34	29	26	23

4 a What is Multicollinearity? What are its causes, consequences, and detection? Discuss. [5] [CO#3]

b A die is thrown 132 times with the following results: [5] [CO#2]

No turned up	1	2	3	4	5	6
frequencies	16	20	25	14	29	28

Test the hypothesis that the die is unbiased
The critical value of chai square at 0.05 level of significance and 5 degree of freedom = 11.07

5 a Explain Classical Linear Regression Model (CLRM) along with its properties. [5] [CO#2]

b The Amrit Merchandising Company wishes to test whether its three salesmen A, B and C tend to make sales of the same order or they differ in their selling ability as measured by the average size of their sales. During the last week, there have been 14 sale calls – A made 5 calls, B and 4 calls and C made 5 calls. Following are the weekly sales record of three salesmen: [5] [CO#2]

A in Rs	B in Rs	C in Rs
300	600	700
400	300	300
300	300	400
500	400	600
0	--	500

Perform the analysis and draw your conclusions. (The critical value of F for v1= 2 and V2=11 degree of freedom at 5% level of significance= = 3.95)

6 a Explain Heteroscedasticity and in what ways can it be detected? [5] [CO#3]

b The Annual sales of a company Excel are as follows: [5] [CO#2]

Year	2015	2016	2017	2018	2019
Sales in Rs	45000	56000	78000	46000	75000

By the least square method, estimate the annual sale for 2025.

7 a What do you mean by Dummy Variable? Discuss with example. In a study of women entrepreneurs, county has been divided in five groups as North, South, East, West and Central. Is it fine to create five dummy variable for the analysis? Discuss with examples in support of your arguments. [5] [CO#3]

b The regression equation of profit (X) and sales (Y) of a firm is $3y-5x+108 = 0$. The average sales of the firm were Rs 44,000 and the variance of the profit is 9/16 of the variance of sales. Find the average profits and coefficient of correlation between sales and profit. [5] [CO#2]